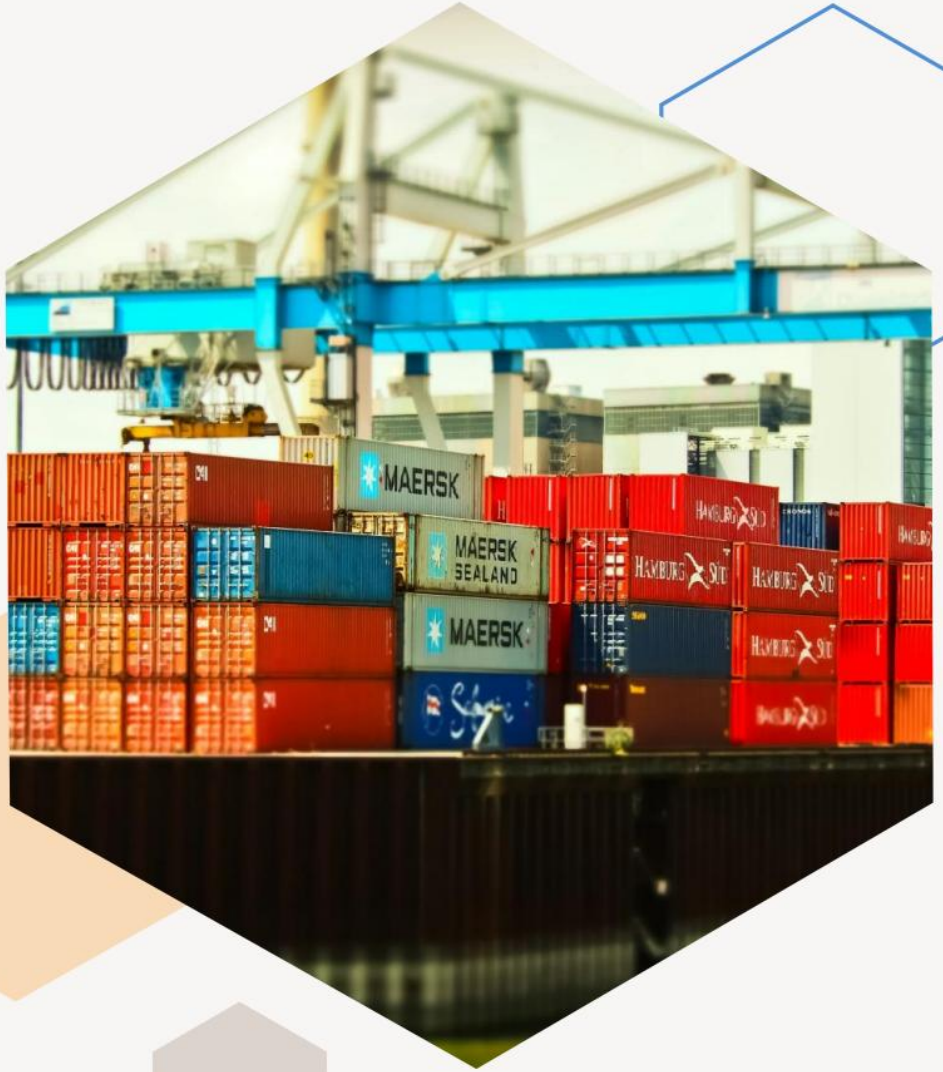
Supply Chain Performance Dashboard & Strategic Analysis 2019

E-Commerce Operations Analytics & Forecasting



- Lamyaa kharmaz

External Analysis - Porter's Five Forces



Competitive Rivalry (HIGH): Major players (Amazon, Alibaba, eBay) compete on price, delivery speed, product range



Supplier Power (MODERATE): Multiple suppliers available, but logistics providers have leverage



Buyer Power (HIGH): Low switching costs, price-sensitive customers, easy comparison shopping



Threat of New Entrants (MODERATE): Low barriers to entry but high cost to scale and compete



Threat of Substitutes (LOW): Traditional retail declining, e-commerce is dominant channel

Data Model & Methodology



Data Model Structure:

Fact Table: FactSales (180,519 transactions with measures: Sales, Profit, Quantity, Discounts)

7 Dimension Tables: DimDate, DimCustomer, DimProduct, DimDepartment, DimShipping, DimCustomerLocation, DimOrderLocation

Relationships: Many-to-One via DateKey, Customer ID, Product ID, Department ID, Location Keys, Shipping Key



Design Rationale:

Star Schema chosen over Snowflake for faster query performance and simpler relationships

Separated customer location vs order location to track billing vs shipping patterns

Date dimension enables time-based analysis (YTD, QoQ growth, forecasting)

Clean data: Removed 86%+ null columns, standardized geographies, converted date formats

Dashboard Design & Analytical Framework



Dashboard Architecture:

Page 1 - Descriptive (What happened?): KPI cards, market performance, category sales, delivery status distribution

Page 2 - Diagnostic (Why did it happen?): Late delivery analysis, profit margins, sales-profit correlation, segment performance

Page 3 - Predictive (What will happen?): Trend analysis with 2019 forecasts, QoQ growth tracking, YTD metrics

Page 4 - Prescriptive (What should we do?): 2019 targets, market-specific goals, strategic recommendations



Chart Selection Rationale:

Bar charts for comparisons (markets, categories) - easy to rank

Line charts with forecasting for trend analysis - shows trajectory

Donut charts for proportional breakdowns - parts of whole

Tables for detailed metrics - precise numbers for decision-making

Gauge for goal tracking - visual target vs actual

Internal Analysis- KPIs & 2019 Objectives

2015–2017

Key Performance Indicators

- Total Sales: $33.05M (peaked 2015: $11.1M, declined to $10.6M by 2017)
- Total Profit: $3.97M (~$1.3M annually sustained despite sales decline)
- Late Deliveries: 54.82% (critical operational weakness)
- Profit Margin: ~11% (stable but needs improvement)

2019

Strategic Objectives

- Sales Target: $15M (41% growth)
- Profit Target: $1.8M (38% increase)
- Delivery Target: <40% late deliveries (15% improvement)
- Target Margin: 12% (margin expansion through pricing optimization)

Sales & Operations Overview

Year: All | Market: All | Category Name: All

Total Revenue (Net Sales)

$33.05M

Number of Customers

20.65K

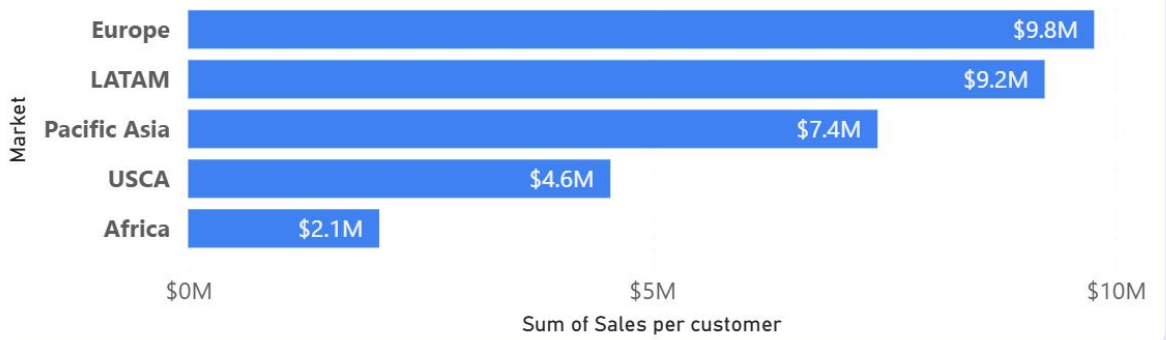
Total Orders

65.75K

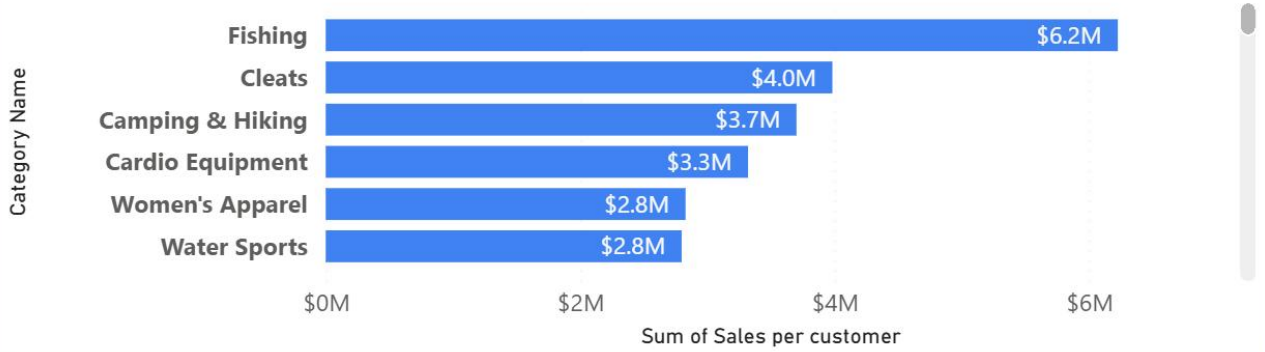
Total Profit

$3.97M

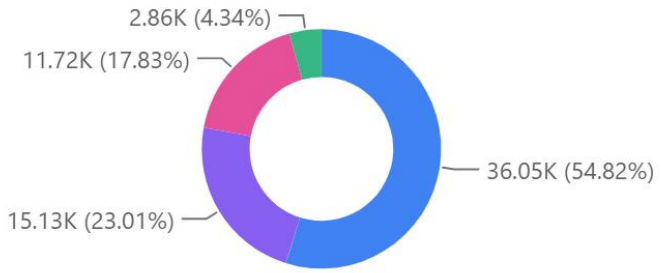
Sales Performance by Market



Top Product Categories



Delivery Status Distribution



Sales by Customer Segment

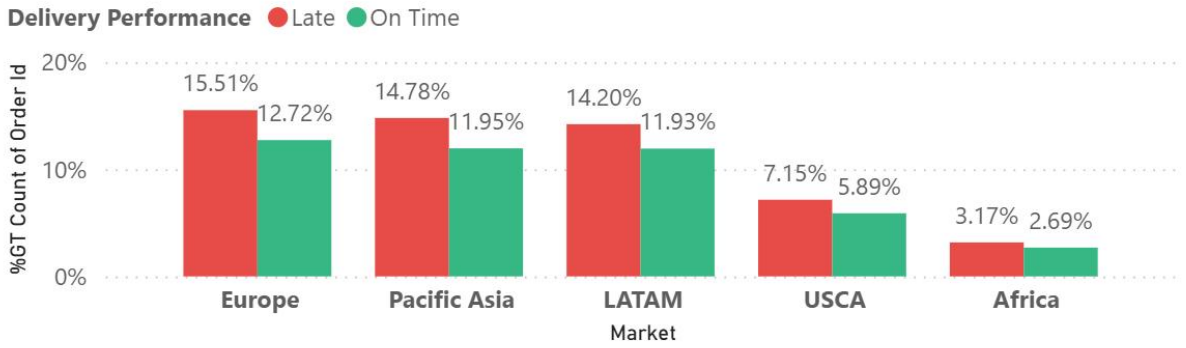


Key Insight:
Europe leads sales ($9.8M) | 54.8% late deliveries need improvement | Consumer segment = 52% of revenue

Performance Analysis

Year: All | Market: All | Category Name: All

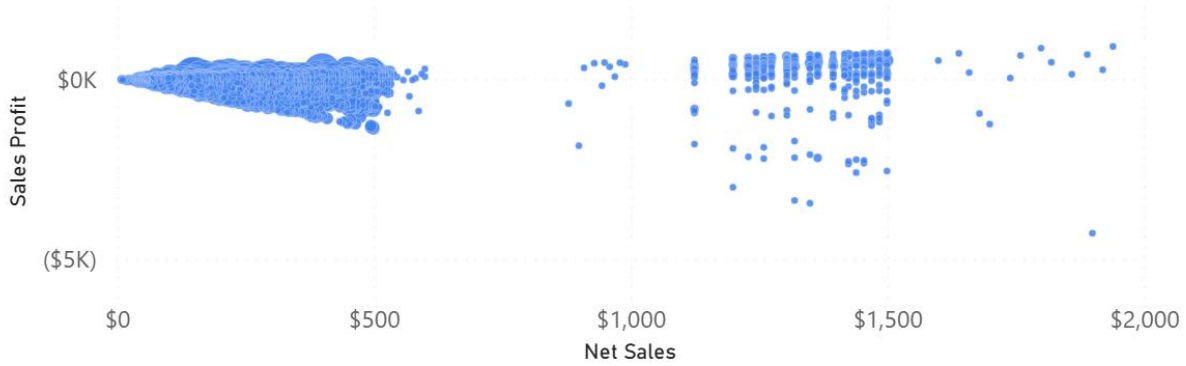
Late Delivery % by Market



Top 10 Late Delivery % by Category



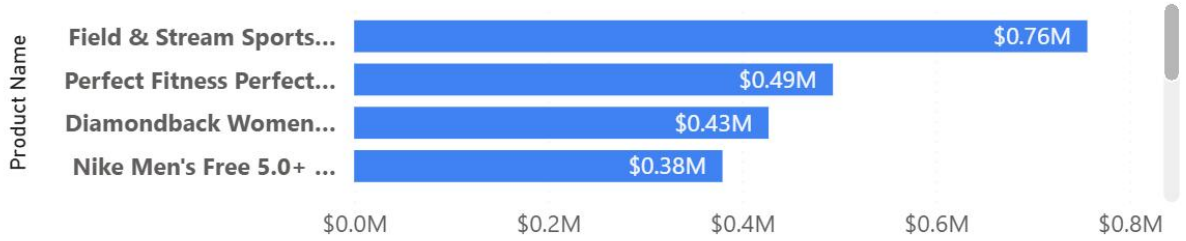
Sales vs Profit Correlation



Top 10 Profit Margin (%) by Category



Top 10 Products by Profit



Customer Segment Performance

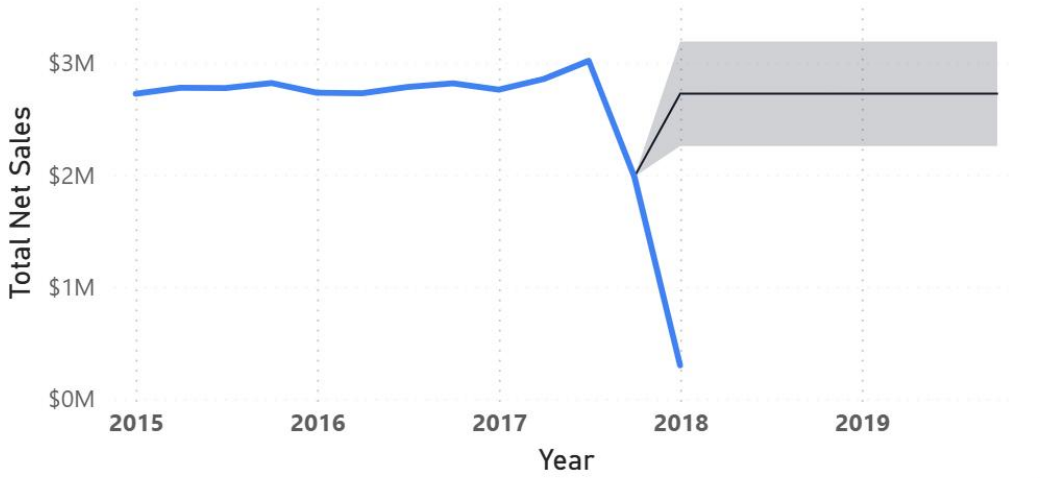
Customer Segment	Sum of Sales per customer	Sum of Order Profit Per Order	Average of Sales pe
Consumer	$17,164,278.37	$2,073,487.67	
Corporate	$10,030,649.90	$1,202,574.96	
Home Office	$5,859,474.11	$690,840.34	
Total	**$33,054,402.38**	**$3,966,902.97**	

Diagnostic Insight: Europe leads late deliveries (15.5%) | High sales = lower margins | Golf Bags most profitable (19%) | Consumer segment dominates revenue

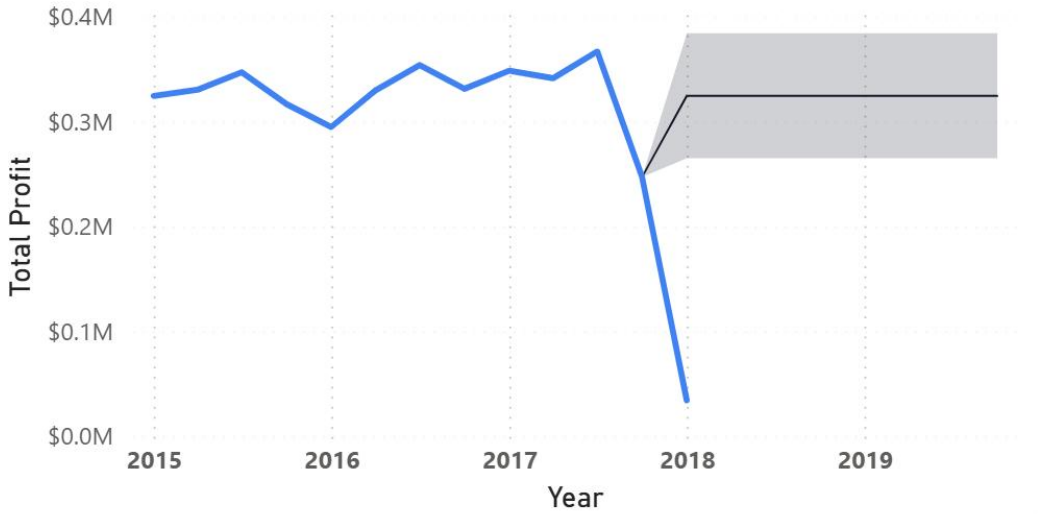
Trends & Forecasting

Year
All

Sales Trend Over Time + 2



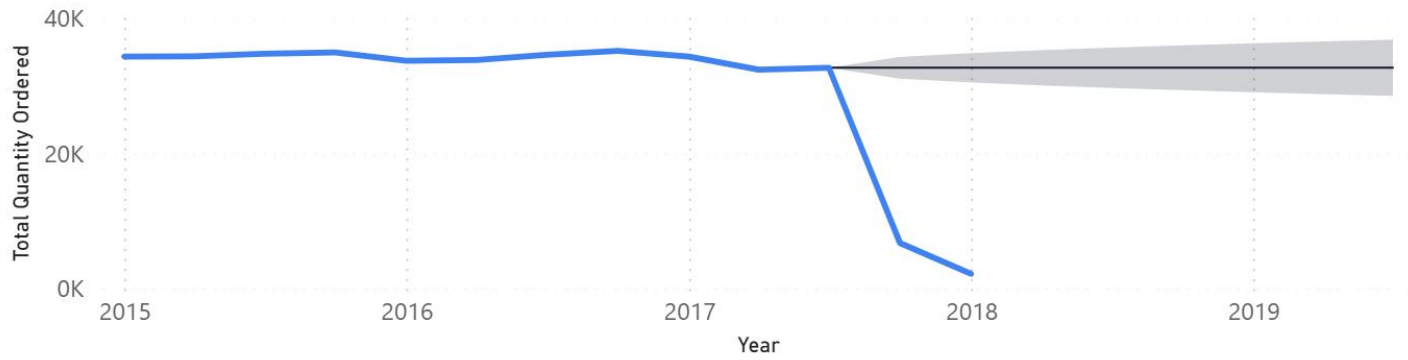
Profit Trend Over Time + 2019 Forecast



Quarterly Performance Analysis

Year	Quarter	Sum of Sales per customer	Previous Quarter Sales	QoQ Growth %	YTD Sales	YTD Profit
2015	Q1	$2,722,444.70			2,722,444.70	324,579.61
2015	Q2	$2,776,186.23	2,722,444.70	1.97	5,498,630.93	654,958.35
2015	Q3	$2,772,652.03	2,776,186.23	-0.13	8,271,282.96	1,002,030.12
2015	Q4	$2,818,260.59	2,772,652.03	1.64	11,089,543.55	1,318,856.90
2016	Q1	$2,731,888.11	2,818,260.59	-3.06	2,731,888.11	294,850.43
2016	Q2	$2,726,919.15	2,731,888.11	-0.18	5,458,807.26	624,739.24
2016	Q3	$2,781,996.97	2,726,919.15	2.02	8,240,804.23	978,795.21
2016	Q4	$2,815,192.53	2,781,996.97	1.19	11,055,996.76	1,310,119.07
2017	Q1	$2,759,687.34	2,815,192.53	-1.97	2,759,687.34	348,643.82
2017	Q2	$2,852,963.22	2,759,687.34	3.38	5,612,650.56	690,018.81
2017	Q3	$3,016,043.04	2,852,963.22	5.72	8,628,693.60	1,057,009.06
2017	Q4	$1,982,216.41	3,016,043.04	-34.28	10,610,910.01	1,304,085.11
2018	Q1	$297,952.05	1,982,216.41	-84.97	297,952.05	33,841.89
Total		**$33,054,402.38**			**297,952.05**	**33,841.89**

Order Volume Trend + 2019 Forecast



Predictive Insight: Sales peaked 2015 ($11.1M), declined to $10.6M by 2017 | Profit sustained at ~$1.3M annually despite sales decline - margin improvement | 2018 incomplete (Q1 only: $298K) | 2019 forecast: ~$2.7M/quarter suggests flat continuation | Q4 2017 dropped 34% QOQ - investigate cause

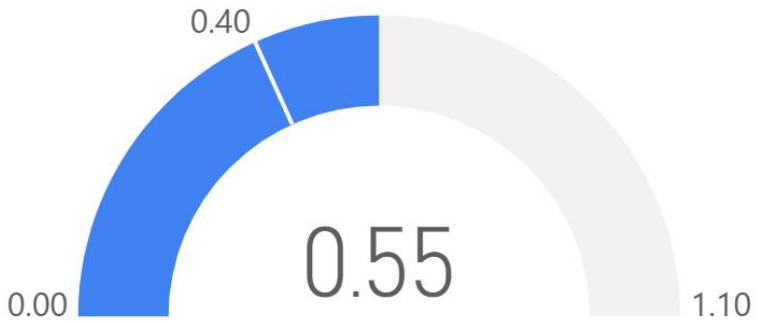
Strategic Goals - 2019 Targets

15.00M
2019 Sales Target

1.80M
2019 Profit Target

12.00
Target Profit Margin %

Delivery Performance Goal



Market Growth Strategy & Targets

Market	2017 Sales	2019 Market Target	Gap to Close	Projected Growth %
Africa	223,838.39	316,426.76	92,588.37	41.36
Europe	4,670,280.72	6,602,092.62	1,931,811.91	41.36
LATAM	4,677,604.50	6,612,445.82	1,934,841.32	41.36
Pacific Asia	1,018,672.00	1,440,034.83	421,362.83	41.36
USCA	20,514.40	28,999.96	8,485.56	41.36
Total	**10,610,910.01**	**15,000,000.00**	**4,389,089.99**	**41.36**

Strategic Recommendations:

Revenue Growth: Ambitious 41% growth target requires aggressive market expansion | **Priority Markets**: Europe & LATAM need $1.9M increase each - largest opportunity | **Operational Excellence**: Reduce late deliveries from 55% to 40% through logistics optimization | **Margin Improvement**: Achieve 12% profit margin (up from 11%) through pricing strategy and cost control | **Quick Wins**: Focus on high-margin categories (Golf Bags 19%, Fitness 17%) for profit acceleration